MOHAMED FAISAL SINDHI

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EDUCATION

University of South Florida, Tampa, FL — Expected Dec. 2026
M.S. in Artificial Intelligence and Business Analytics · GPA 3.86/4.00

Anna University, College of Engineering, Chennai, India — Nov. 2020
B.E. in Computer Science and Engineering

RESEARCH EXPERIENCE

Tracing Recommendations Across Clinical Protocol Editions — Aug. 2026–Present
Lead Researcher, RAEY
Collaborator: Dr. Nasir Uddin

- Asked whether each recommendation in a state EMS clinical protocol can be traced to its counterpart in the next edition. I prespecified hypotheses and analyses and froze the method before retrieving test documents.
- My method kept more recommendations connected across editions than matching by identifier alone, but a simpler text matcher did just as well. Recommendations that moved between sections remained the problem I most want to solve.

  Analysis plan, full results, and reproducibility package: [link]

University of South Florida, Tampa, FL — Jan. 2025–Jan. 2026; Aug. 2026–Present
Research Assistant
Advisors: Dr. Ehsan Sheybani and Dr. Arash Takshi

- **Aug. 2026–Present:** Exploring DOpO2 and dissolved oxygen measurement using spectrometers (early stage).
- **Jan. 2025–Jan. 2026:** Helped clean photoluminescence spectra of Mg-doped ZnO phosphors and ran PCA and Random Forest classification (80/20 split, 3-fold CV); two components explained over 95% of variance and the best model reached about 0.72 mean cross-validation accuracy. Summarized the results for a lab presentation.

PREPRINTS AND MANUSCRIPTS

Preprints

P3 **When Verification Passes the Wrong Calculator: A Human-Adjudicated Study of a Guarded Local LLM Pipeline**
M.F. Sindhi, N. Uddin
Research Square, 2026 [link]

P2 **Do Artificial Masks Represent Natural Missingness? External-Reference Evaluation in County-Level Power-Outage Records**
M.F. Sindhi
Zenodo, 2026 [link]

P1 **Validation of a Frozen LLM-Based Screening Instrument for Clinical Guideline Literature Screening**
M.F. Sindhi
Zenodo, 2026 · earlier version on SSRN [link]

Submitted Manuscripts (public records pending)

S2 **Approximate Sampling for Randomization Tests: A Total-Variation Certification Framework**
M.F. Sindhi, H. Buksh
Submitted, 2026

S1 **Bandwidth, Representation Loss, and Temporal Drift in Quantum Kernel Fraud Detection: A Pre-Specified Falsification Study**
M.F. Sindhi, H. Buksh
Submitted, 2026

Applied AI Projects

AccessPath: Accessible Routing with Uncertain Data 2026
Python, FastAPI, PostGIS, NetworkX, React · Code

- Built a Seattle route planner that treats missing accessibility data as unknown rather than safe; after an adversarial test broke my first scoring rule, I redesigned it so one reliable hazard report cannot be outvoted, and on a small fixed benchmark the router then avoided unknown segments where possible.

F-1 Policy Intelligence: Policy Q&A with Citations 2026
Next.js, TypeScript, PostgreSQL, pgvector, Gemini · Code

- As an international student who wanted quick answers to visa questions, I built a USF version of a tool another student made, which answers F-1 questions from federal regulation and USF guidance, cites every claim, and declines when the evidence is weak; results on a small development set of 28 questions are preliminary.

SENTINEL: Patient Deterioration Warning 2026
Python, GPT-4o, Llama 3.1, Med42-v2

- Built an early warning system for patient deterioration with six agents and compared three LLMs with a rule-based baseline on synthetic patients: the system flagged deterioration about 2.7 hours earlier than NEWS2, a standard hospital warning score, and the rule-based baseline beat the LLMs.

PlateProof: Restaurant Inspection Intelligence 2026
Python, FastAPI, DuckDB, scikit-learn · Code

- Working alone, I built a lookup tool for NYC and Florida restaurants showing each one's official inspection history and a calibrated risk estimate for its next inspection, keeping each jurisdiction's grading system and shipping a model only if it beats baselines.

Teaching Experience

University of South Florida, Muma College of Business, Tampa, FL May 2026–Present
Graduate Teaching Assistant
Supervisor: Dr. Nasir Uddin

- AI & Analytics (CAI 3801): one section in Summer 2026 and three in Fall 2026.
- Data Analytics for Business (QMB 3302): Summer 2026.
- Assisting with instruction and grading, and helping students with Power BI and Tableau, across sections with combined enrollment above 500.

Entrepreneurial Projects

RAEY, Tampa, FL Aug. 2026–Present
Founder (prelaunch venture)

- Built a hospital SOP prototype (Next.js, FastAPI, SQLite, Docker) that answers only when it can cite a supporting SOP passage; tested on synthetic SOPs only, not clinically validated.
- Met several times with a doctor at Moffitt Cancer Center to explain the problem and use case and to discuss how RAEY would need to change for real physician use.

Solely Active, Remote Sep. 2026–Present
Technical Product Lead (prelaunch venture)

- PlateProof, my independent project, was taken up by Solely Active, a sports and wellness venture. I lead technology and product direction for its Active Twin concept and coordinate an international team of three exploring food recognition, recommendations, and careful handling of personal data.

Professional Experience

USF Information Technology, Tampa, FL — Dec. 2025–Present
Microsoft Student Ambassador Intern

- After training from Microsoft in Copilot and Microsoft 365, built a workflow for Graduate Advising that ended duplicate record entry and saves staff 3 to 4 hours a month; now supporting the Office of Resource Management and the College of Medicine on projects still in progress.

Vee4 Software, United Arab Emirates — Sep. 2022–Nov. 2024
Blockchain Engineer (C++/Python)

- Shipped production blockchain and name service features in C++ and Python, and implemented homomorphic encryption and federated learning workflows so models learned without pooling client data.

TechResx Technologies, Chennai, India — Mar. 2021–Mar. 2022
SEO Analyst and WordPress Developer

- Developed PHP/WordPress sites with API integrations; my SEO and Google Ads analysis raised traffic 20% and average session duration 25% within three months.

United National Dairy Company (Rayan), Manama, Bahrain — May 2019–Jul. 2019
Sales Operations Analyst Intern

- Traced supply chain workflows from manufacturing to retail, identified process gaps, and recommended branch improvements; also supported daily operations.

Technical Skills

Programming: Python, SQL, C++, TypeScript, R

Machine Learning and Evaluation: PyTorch, scikit-learn, dimensionality reduction, ensemble methods, cross-validation, model calibration

Language Models and Retrieval: retrieval-augmented generation, hybrid retrieval, reranking, citation verification, source provenance, abstention

Data and Software Tools: PostgreSQL, pgvector, PostGIS, DuckDB, FastAPI, Docker, Git, GitHub Actions

Additional Training and Certifications

- Probability for AI, Stanford University — In progress
- GCI World 2026, Matsuo-Iwasawa Lab, University of Tokyo — In progress
- AI Engineer for Data Scientists Associate, DataCamp — Sep. 2026
- SnowPro Associate: Platform, Snowflake — Dec. 2025
- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional, Oracle — Aug. 2025